

KASTSYUKOVICHY, Belarus

WMO: 268870

Lat: 53.3456N

Lon: 32.0738E

Elev: 167

StdP: 99.33

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-18.9	-24.8	0.4	-21.3	-21.9	0.5	-18.6	11.2	-3.4	10.0	-2.5	2.2	70	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.4	20.3	27.5	19.4	25.9	18.6	21.3	27.3	20.4	26.0	19.4	24.7	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.3	24.4	18.3	13.5	23.1	17.4	12.7	22.1	62.6	27.5	59.2	26.1	56.1	24.7	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.8	7.7	DB	-26.1	31.7	2.7	2.4	-28.1	33.4	-29.7	34.8	-31.2	36.2	-33.1	38.0	(4)
(5)			WB	-26.3	23.1	2.7	1.7	-28.3	24.3	-29.8	25.3	-31.3	26.3	-33.2	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-5.7	-5.1	-0.1	7.7	13.1	16.6	19.1	17.5	12.2	6.2	0.9	-3.7	(6)	
	DBStd	9.94	6.19	6.36	4.71	4.32	4.41	3.46	3.05	3.40	3.63	4.05	5.08	5.92	(7)	
	HDD10.0	2186	487	423	314	94	19	1	0	0	19	130	274	426	(8)	
	HDD18.3	4385	745	657	572	318	169	73	27	56	185	376	523	684	(9)	
	CDD10.0	958	0	0	1	26	116	200	283	234	86	13	1	0	(10)	
	CDD18.3	115	0	0	0	8	23	51	32	2	0	0	0	0	(11)	
	CDH23.3	1081	0	0	0	4	80	188	457	324	27	1	0	0	(12)	
	CDH26.7	274	0	0	0	1	11	28	127	106	2	0	0	0	(13)	
(14) Wind	WSAvg	3.5	3.7	3.7	3.9	3.7	3.5	3.2	2.9	3.1	3.1	3.6	3.7	3.6	(14)	
(15) Precipitation	PrecAvg	651	39	37	35	40	64	75	83	60	61	64	49	42	(15)	
	PrecMax	844	75	70	70	97	133	169	241	154	132	136	90	92	(16)	
	PrecMin	457	10	16	5	12	14	19	21	11	8	7	6	14	(17)	
	PrecStd	99	17	14	16	20	24	33	48	33	30	37	22	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	6.2	14.7	23.0	28.1	29.2	32.6	35.0	26.4	20.5	12.6	8.6	(19)	
		MCWB	4.7	4.0	8.1	13.2	19.1	19.9	21.9	20.1	16.4	13.6	11.1	7.7	(20)	
	2%	DB	3.6	4.0	11.1	20.0	25.7	27.5	29.9	29.1	23.3	16.5	9.8	6.4	(21)	
		MCWB	2.6	2.8	6.1	11.8	17.5	19.5	20.6	20.4	16.0	13.1	8.7	5.5	(22)	
	5%	DB	2.3	2.8	8.4	17.7	23.4	25.8	27.7	26.6	20.6	14.0	8.4	4.2	(23)	
		MCWB	1.5	1.8	4.9	10.9	15.9	18.8	19.8	18.8	14.8	11.4	7.5	3.6	(24)	
	10%	DB	1.2	1.8	6.1	15.3	21.2	23.8	25.9	24.4	18.5	12.1	7.1	2.6	(25)	
		MCWB	0.5	1.0	3.6	9.8	14.7	17.5	19.1	17.7	13.8	10.1	6.2	1.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	4.8	9.0	14.3	20.3	21.2	22.7	22.2	18.1	15.8	11.3	8.0	(27)	
		MCDB	5.5	5.7	14.0	20.7	26.6	26.8	29.4	27.8	24.5	18.9	12.4	8.6	(28)	
	2%	WB	2.8	3.0	6.8	12.7	18.3	20.3	21.7	21.0	16.7	13.5	9.0	5.6	(29)	
		MCDB	3.4	3.7	10.4	17.9	23.8	26.1	28.1	27.3	21.4	15.8	9.7	6.2	(30)	
	5%	WB	1.6	2.0	5.2	11.6	17.0	19.4	20.6	19.9	15.6	11.7	7.6	3.7	(31)	
		MCDB	2.2	2.7	7.9	16.4	22.4	24.7	25.9	26.0	19.1	13.6	8.3	4.2	(32)	
	10%	WB	0.6	1.1	3.7	10.4	15.6	18.3	19.6	18.5	14.4	10.5	6.3	2.0	(33)	
		MCDB	1.2	1.7	5.8	14.6	20.2	23.0	24.6	23.1	17.3	12.0	7.0	2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.6	6.8	9.2	10.2	9.8	10.2	10.7	9.5	6.9	4.2	4.1	(35)	
		MCDBR	3.3	5.0	10.2	13.0	13.4	12.4	13.3	14.3	13.1	10.2	5.5	4.2	(36)	
	5% WB	MCWBR	3.3	4.3	6.1	6.4	6.5	5.9	5.6	6.0	6.7	6.6	4.5	3.8	(37)	
		MCWBR	3.2	4.6	9.0	11.3	12.2	11.3	11.7	12.7	11.1	8.9	4.7	4.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.294	0.328	0.356	0.421	0.398	0.383	0.415	0.425	0.380	0.354	0.334	0.298	(40)	
		taud	2.132	2.070	2.123	2.143	2.295	2.357	2.288	2.245	2.337	2.374	2.257	2.160	(41)	
	Edn at Noon		650	742	811	805	851	867	829	794	787	726	605	567	(42)	
		Edh at Noon	69	102	120	135	122	116	122	121	98	76	63	57	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.73	1.63	3.04	4.07	5.42	5.77	5.54	4.66	3.11	1.56	0.59	0.44	(44)	
	RadStd		0.15	0.28	0.44	0.42	0.37	0.49	0.48	0.46	0.31	0.30	0.12	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.23	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.62	N/A	N/A	N/A	N/A	N/A	-122	-197	+94	N/A

Nomenclature: See separate page